

A Comparative Evaluation of Heterogeneous Ensemble Voting and Stacking Strategies for Multi-Class Flow-Based DDoS Detection

By

Dan Kinsiona

BSc Cyber Security

Supervisor: John Heggarty

Sheffield Hallam University

This student has a registered Learning Contract with [Disabled Student Support]. Please mark this assignment for content and ideas only, disregarding errors in spelling, punctuation, and grammar, as recommended in their contract.

TABLE OF CONTENTS

Abstract.....	5
1. Introduction.....	6
2. Literature Review.....	8
3. Research Methodology.....	12
3.1 Research design and approach.....	12
3.2 Dataset generation and feature extraction.....	13
3.3 Classifiers and ensemble strategies.....	14
3.4 Experimental setup and cross-validation.....	16
3.5 Evaluation metrics, statistics and implementation.....	16
3.6 Ethics.....	17
4. Results.....	18
4.1 Baseline classifier performance.....	18
4.2 Per-class results (Ensemble).....	19
4.3 Feature reduction impact (Run A vs Run B).....	21
4.4 Scaling impact (Run C and Run D).....	24
5. Discussion.....	25
5.1 Feature-reduction tradeoff and heterogeneity collapse.....	25
5.2 Scale-only validation and stability over victory (Run C).....	26
5.3 Statistical confirmation via McNemar test.....	27
5.4 Production-scale validation (Run B vs Run D).....	28
6. Conclusions.....	29
6.1 Findings.....	29
6.2 Recommendations.....	29
6.3 Limitations.....	30
6.4 Future Work.....	30
6.5 Contributions.....	31
Acknowledgements.....	32
References.....	33

Appendices.....	36
Appendix A. Full per-class table	36
Appendix B. Per-Fold Metrics for All Strategies and Runs	38
Appendix C. Baseline and Ensemble Leaderboards at Full Precision.....	39
Appendix D. Per-Fold Disagreement Across All Four Runs and Strategies	42
Appendix E. AI Transparency Statement (AITS 2)	42
Appendix F. Git Hub Repository.....	43
Appendix G: Project Proposal	Error! Bookmark not defined.

List Of Tables

TABLE 1 ATTACK TYPES	13
TABLE 2: BASELINE LEADERBOARD	18
TABLE 3: ENSEMBLE STRATEGY LEADERBOARD	19
TABLE 4: CROSS-RUN STRATEGY DELTA - RUN A VS RUN B	21
TABLE 5: AVERAGE PAIRWISE DISAGREEMENT PER-FOLD, RUN A VS RUN B	22
TABLE 6: CROSS-RUN ENSEMBLE COMPARISON ALL METRICS, ALL FOUR RUNS	24
TABLE 7: APPENDIX A - FULL FEATURE LIST.....	36
TABLE 8:APPENDIX B - PER-FOLD F1 AND AGGREGATE METRICS, RUN A (27 FEATURES, 99,999 FLOWS)	38
TABLE 9 APPENDIX B - PER-FOLD F1 AND AGGREGATE METRICS, RUN B (22 FEATURES, 99,999 FLOWS)	38
TABLE 10: APPENDIX B - PER-FOLD F1 AND AGGREGATE METRICS, RUN C (27 FEATURES, SCALED SUBSET)	38
TABLE 11: APPENDIX B - PER-FOLD F1 AND AGGREGATE METRICS, RUN D (22 FEATURES, SCALED SUBSET)	39
TABLE 12: APPENDIX C - BASELINE CLASSIFIER RANKING FOR RUN A BY WEIGHTED PERFORMANCE SCORE (WPS).....	39
TABLE 13: APPENDIX C - BASELINE CLASSIFIER RANKING FOR RUN B AFTER FEATURE REDUCTION (22 FEATURES).....	40
TABLE 14: APPENDIX C - ENSEMBLE STRATEGY LEADERBOARD, RUN A (27 FEATURES, 99,999 FLOWS)	40
TABLE 15: APPENDIX C - ENSEMBLE STRATEGY LEADERBOARD, RUN B (22 FEATURES, 99,999 FLOWS)	41
TABLE 16: APPENDIX C - ENSEMBLE STRATEGY LEADERBOARD, RUN C (27 FEATURES, SCALED SUBSET)	41
TABLE 17: APPENDIX C - ENSEMBLE STRATEGY LEADERBOARD, RUN D (22 FEATURES, SCALED SUBSET)	41
TABLE 18: APPENDIX D - PAIRWISE DISAGREEMENT (%) ACROSS FIVE CROSS-VALIDATION FOLDS FOR ALL FOUR EXPERIMENTAL.....	42

Table Of Figures

FIGURE 1 END-TO-END CONSTRUCTION PIPELINE FROM TRAFFIC GENERATION TO BALANCED DATASET:.....	13
FIGURE 2: THREE STAGE SELECTION PROCESS. BASELINE TO ENSEMBLE PIPELINE	15
FIGURE 3: FORMULA AND METRIC WEIGHTS.	17
FIGURE 4: PER-CLASS CONFUSION MATRIX AND BAR-CHART PERFORMANCE FOR MAJORITY.....	20
FIGURE 5: PAIRWISE DISAGREEMENT BAR-CHART RUN A VS RUN B	22
FIGURE 6: PER-CLASS CONFUSION MATRIX AND BAR-CHART.....	23
FIGURE 7:MCNEMAR'S TEST	27
FIGURE 8: WHAT DRIVES DISAGREEMENT? - OVERVIEW BAR-CHART OR ALL RUNS	28
FIGURE 9: FULL FEATURE DESCRIPTION LIST	37

ABSTRACT

DDoS attacks have risen in sophistication, pursuing the use of ensemble techniques over single machine learning models that do not have the learning diversity for nonlinear traffic. Multi-class flow-based DDoS detection through heterogeneous ensemble comparison remains underexplored, with limited head-to-head evaluation under strict conditions. This study addresses the question. 'Which heterogeneous ensemble strategy is most effective for flow based multi class DDoS detection?' We hypothesise that no single strategy will dominate across all metrics and that stability over victory will determine which is most appropriate. Nine balanced traffic classes were generated from hping3 for eight attack types and iperf for benign traffic. Four runs (A, B, C, D) compared Majority Voting, Weighted Voting, Adaptive Voting and Stacking, with base classifiers selected using a weighted performance score (supporting per-fold data in Appendix B). Run A revealed the collapse of heterogeneity at 0.0% disagreement (per-fold disagreement in Appendix D). Run B forces the disagreement to 39.01% through feature reductions by removing TCP flags characterised to DDoS, Stacking led on Run B's reduced test with macro-F1=0.4219 and maintained scores of 0.9634 on previous runs. Run C at 499,995 flows held disagreement near zero (0.0126%) with Stacking F1 std 0.000235 vs voters 0.000307 (24% less variance). Run D held disagreement at 37.78% at the same scale. Stacking emerges as most effective on stability, making stability over victory a deployable principle for production environments handling nonlinear traffic.

1. INTRODUCTION

Records show DDoS attacks have impacted industry leading organisations throughout history, including GitHub (March 2015), Google Cloud (September 2017) and Azure (2020, 2021) (Cloudflare, n.d.). The rate of DDoS attacks has risen drastically over the years. Cloudflare mitigated 47.1 million DDoS attacks in 2025, which is double the previous year, averaging 5,376 attacks every hour (Cloudflare, 2026a). Cloudflare's 2026 Threat Report tells us that the automation of attacker operations through AI is a defining trend (Cloudflare, 2026b). Denial of Service (DoS) is a cyber-attack from a single source that overwhelms a server or network with traffic. Distributed Denial of Service (DDoS) is considered superior, deriving from DoS but attacks are more intensified from multiple sources simultaneously. These attacks are coordinated by a single host communicating with devices on a global scale. Malicious software is distributed through infected sites or alternative vectors that recruit devices into botnets through a centralised command and control strategy. This forces security infrastructure to develop in the areas of network intrusion and detection, as the impact on businesses can be severe, causing financial, regulatory and reputational damage by compromising availability and overloading system constraints such as CPU and network bandwidth.

Machine learning supports DDoS detection by learning patterns from network traffic by classifying the data between normal traffic and attack type classes, acting as an active improvement to traditional rule and signature-based detection methods which may include human error and knowledge-based limitations. Base models are single classifiers which operate individually based on their algorithms learning capabilities which can still be effective with linear traffic however complexity within nonlinear traffic requires a diverse approach this is where Ensemble learning takes designs further by combining multiple base classifiers into a single decision, which Dietterich, (2000) shows reduces the risk of misclassification compared to any one model. The ensemble strategies compared in this study are Majority Voting, Weighted Voting, Stacking and Adaptive Voting.

Recent research covers ensemble strategies against single model performances across a range of conditions. The problem statement that this study addresses is that head-to-head comparison of ensemble strategies under strict conditions remains limited, restricting practical recommendations and ensemble selection per use case. Therefore, this study is structured towards DDoS detection.

The primary objective addresses the gap:

- **Primary:** Which heterogeneous ensemble strategy is most effective for multi-class DDoS detection?

To achieve this the paper makes the following major contributions:

1. **Maintained a single controlled experimental environment** using identical datasets, preprocessing and evaluating conditions across strategy comparisons to produce more consistent and comparable conclusions than those drawn from existing literature where conditions vary across studies.
2. **Evaluated the contribution of each base classifier** to the overall ensemble performance under fixed experimental conditions to establish benchmark performance.
3. **Conducted a head-to-head comparison** of Majority Voting, Weighted Voting, Adaptive Voting and Stacking strategies to identify effective mechanisms for multi class DDoS traffic.
4. **Assessed practical recommendations** for DDoS detection and reliability through the evaluation of results.
5. **Tested the effects of feature reduction on ensemble heterogeneity** by introducing Run B to bring forward diversity and disagreement amongst strategy behaviour to drive margins between closely performing scores.
6. **Implemented scalability methods to stress test ensemble strategies** by increasing dataset volume for Run C and D.

We hypothesise that no single ensemble strategy will have complete dominance across all metrics and that stability over victory will determine which strategy is most appropriate. Heterogeneous ensemble strategy selection for production environments depends on stability over victory and generalisation capability to nonlinear traffic, which this study investigates through a head-to-head comparison.

Moving forward, this research paper is structured into six main sections followed by five appendices which display full run results and further methodology details. Section 2 delivers the literature review that analyses current and past research on DDoS detection and identifies the specific gaps this study aims to address. Section 3 describes the research methodology, including the rationale for the quantitative design, dataset generation through traffic simulation and the technical configuration of the four ensemble strategies. Section 4 reveals the results across the four runs A, B, C and D. Section 5 discusses the findings against the research question and the existing literature. Section 6 closes the research with the conclusion of key findings, limitations, practical recommendations, future work and contributions towards network security.

2. LITERATURE REVIEW

The Literature Review examines ensemble machine learning methods for DDoS detection through Voting strategies in cybersecurity. Recent research demonstrates that ensemble approaches can outperform individual classifiers for DDoS attack detection, with Hasanah et al. (2024) described the impact of their weighted voting experiment that scored the highest precision of 98.66% on the UNSW-NB15 dataset. Verma et al. (2024) provided evidence of Stacking achieving accuracy almost close to 99.9% on CICIDS-2017. The work of Angulo et al. (2026) offer evidence that a Stacking and Voting ensemble, on a nine-class CIC-DDoS2019 configuration, made direct comparisons to the dataset structure used in this study, achieved an accuracy of 91.26% and macro F1-scores greater than 0.90. However, performance metrics change across studies depending on methodology approach which also relates to the selection process of base models, ensemble strategies and dataset attributes. This range of variables show that ensemble performance is sensitive to the design choices. Constructing heterogeneous ensembles mean incorporating different base classifiers that offer complementary decision making. Which can reduce the individual limitations of a model by improving overall generalisation capability. Acharya et al. (2021) tested this by combining several base classifiers on NSL-KDD, KDD99 and UNSW-NB15, where the heterogeneous ensemble reduced the limitations of standalone classifiers across binary and multi-class intrusion detection. The foundational work of Dietterich (2000) shows that there are three main points of justification for this, the first reason is statistical. Baseline models choose one classifier from the hypothesis space while ensembles choose multiple and then average the predictions which reduces the statistical risk of misclassification. The second reason is computational, where single classifiers can get stuck in local optima due to a single starting point. Ensembles combine multiple learners with different starting points to avoid this improving the chances of reaching the global optimum, the closest form of correctness. The third reason is representational. Algorithms partition data by drawing decision boundaries and no single algorithm can represent every possible boundary shape in real data. Ensembles draw multiple boundaries through different learner families. For example, decision trees cannot be drawn diagonally, so approximation from different angles is used to separate the data. The use of multiple learning mechanisms in the ensemble improves DDoS detection performance through linear and nonlinear pattern recognition. DDoS attacks vary across protocols (TCP, UDP, ICMP) and traffic intensity (Cloudflare, 2026b), therefore no single model can detect all forms consistently which is why heterogeneity is necessary. Prior work from Oyucu et al. (2023) has demonstrated that ensembles can prove to be more accurate with prediction and work on multi-protocol dataset classes (TCP, UDP, ICMP) using hping3 generated traffic, the same toolkit used here, which establishes a methodological approach for this research within DDoS detection.

While differences between base classifier contributes to ensemble robustness the method used to combine individual model predictions significantly influences overall detection performance. Majority Hard Voting gives each base classifier equal weight. The class with the highest votes wins the final prediction. Allowing computational efficiency. However, this method can be less reliable than weighted alternatives as author Hasanah et al. (2024) explains that this type of ensemble can be improved by assigning weights, since stronger base classifiers within the ensemble can often lose out on votes regardless of producing correct predictions as weaker one's vote may count equally if decided by majority. Weighted Voting provides solution to this issue through weights assigned to base models that outperform its counterparts which will receive dominance over them when predictions are made (Hasanah et al., 2024). However, due to the application of static logic, the performance of the system continuously relies on the weights, because the static weights cannot adapt to the shifts in base model performance or traffic patterns (Alashhab et al., 2024). Therefore, in these instances, the system will need to go through a re-configuration to sustain the performance. Demonstrated by the author Verma et al. (2024), Stacking introduces a meta-learning approach in which the predictions generated by the base classifiers are used as inputs to a secondary model, which is trained to produce the final classification outcome. This allows the meta-learner to learn relationships between base classifier outputs without predefined weighting schemes. But can increase computational complexity as multiple base classifiers must be trained before their outputs are processed by an additional model. Stacking also increases the risk of overfitting where the meta-learner becomes tailored to the training data, reducing generalisation to unseen attack patterns. Angulo et al. (2026) demonstrated that Stacking models can prove to be robust when the meta learner's risk of over fitting the training data is measured. Jiyad et al. (2024) found that hyperparameter tuning reduces overfitting in ensemble models for DDoS detection, where their XGBoost ensemble reached 97% accuracy and 97% F1. Moving on from Stacking, Adaptive Voting addresses the limitations of static weights with the use of dynamic weights based on monitored performance metrics, allowing higher performing classifiers to gain more priority as performance conditions evolve. However, as Mulimani and Patil (2026) noted, Adaptive Voting's performs when traffic and conditions are live and evolving, so that dynamic weight adjustments can respond to the shifting performance in real time which is the reason why static dataset traffic does not completely capture the strengths. Adaptive Voting was still tested because it gives insight into how effective it is on static traffic, as adaptability is a feature of the strategy rather than the only measure of performance. These differences across Voting and Stacking strategies set the basis for head-to-head comparison under identical conditions in this study, where structural simplicity, weighting and meta-learning are tested directly against one another.

Moving beyond ensemble construction and prediction combination strategies, reported performance is also shaped by dataset selection and evaluation methodology. Studies frequently use different benchmark datasets such as UNSW NB15 (Hasanah et al., 2024) which vary in traffic distribution, attack diversity and class imbalance, furthermore preprocessing techniques including normalisation, class balancing methods such as Smote differ across studies. Feature selection approaches such as ANOVA F-value used by Hasanah et al. (2024) also differ across studies. The work of Kim et al. (2025) shows that class balancing methods such as over and under sampling can change reported performance outcomes in DDoS classification. In addition to dataset attributes, differences in reported performance metrics further create difficulties for cross study comparisons as some studies prioritise accuracy while others prioritise precision, F1 macro score or the other common metrics as accuracy alone may be inaccurate on occasions when which it is measured against imbalanced datasets where highly prioritising the overall use of accuracy can mask false positive or false negative rates (Hancock et al., 2023). The work of Grandini et al. (2020) shows that multi-class metrics each respond differently to class imbalance. There can be a lot of differences across the studies in terms of validation methods, although historically, they have used either training-test splits or K-fold methods. A train-test split divides a data set into a single training set and a single testing set. K-fold creates multiple folds so that every part of the data set is used as training and testing data. The work of Abedin et al. (2026) shows that changing k number of folds can make results less stable and affect the comparison of performance metrics across studies. Across the reviewed literature, the combination of head-to-head comparison of four ensemble strategies (Majority Voting, Weighted Voting, Stacking and Adaptive Voting), a nine-class dataset generated from hping3, identical conditions (dataset, preprocessing and evaluation) and stability of ensemble strategy across runs has not been jointly evaluated within a single experimental pipeline. Kim et al. (2025) and Hancock et al. (2023) take different positions on imbalanced data, where Kim shows balancing changes reported outcomes and Hancock shows accuracy alone may be misleading on imbalanced datasets. This study addresses both points by balancing the dataset so no class has an advantage, while adapting the weighted performance scoring approach of Pandey et al. (2024) to downgrade accuracies influence on predictions and give more weight to MCC and F1-Macro which will contribute to predictions more effectively.

The gap analysis has highlighted the main space which will be addressed in this systematic comparative evaluation of ensemble strategies which remains limited, as the survey by Alharthi et al. (2025) suggests across the multi-class IDS work taken place. From this reviewed area we have discovered three specific research gaps. The first, head-to-head comparison in ensemble strategies serving as the primary variable of investigation which is limited within recent and previous literatures. Maodah et al. (2025) recently compared Hard Voting against Stacking on a single dataset which leaves room for the comparison within experiments that include a wider range of ensemble designs. The second gap takes a look at reliability in the sense of how stable can the ensemble strategies perform across multi-runs when selective conditions change which is limited in current reports across modern timelines with developed technology falling short in the wider area of ensemble work, as Cheng and Feng (2026) confirmed when comparing majority voting without comparing against other ensemble strategies. The third gap which links the previous two gaps is directed towards industry standards and clear guidance on ensemble strategy selection for practitioners or per use case within production environments and in relation to high availability and DDoS detection which remains underdeveloped, which has also been mentioned as an issue by the author Lucas et al. (2023) conducted a systematic literature review analysing 256 papers which has been filtered to a total of 188 papers across six databases (2015–2022) as a final study, following the SLR protocol which follows techniques such as bias reduction and filtering, which reinforces the gap.

3. RESEARCH METHODOLOGY

Here we outline the methodology used for the detection of flow-based DDoS attacks through supervised machine learning. The methodology is based on the head-to-head comparison of Majority Voting, Weighted Voting, Adaptive Voting and Stacking under identical conditions, to evaluate stability over victory.

3.1 Research design and approach

This research aligned with the Saunders' research onion (Mvalo, 2026; Saunders et al., 2019) as a positivist, deductive study using a quantitative methodological choice, an experimental strategy with a cross-sectional time horizon, and techniques including 5-fold cross-validation, hyperparameter tuning and McNemar's statistical testing.

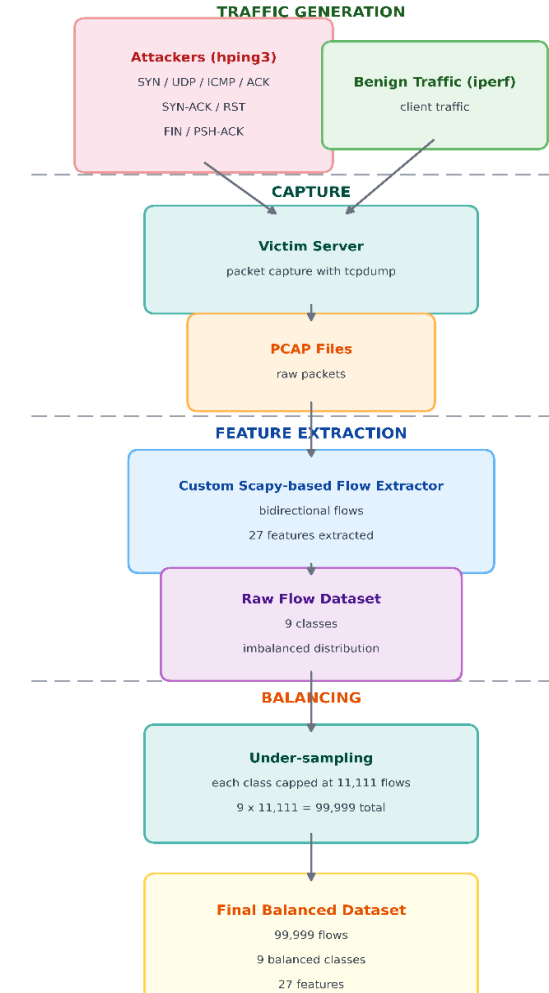
This quantitative approach is applied through a nine-class classification task benign traffic and eight DDoS attack types to produce realistic network monitoring by identifying different attack behaviours. All datasets were created and balanced in a controlled environment using under-sampling to equally size all nine classes. The first dataset used 27 flow features to capture the main indicators of DDoS activity. The second dataset reduces the features to 22 by removing TCP flags to test the ensembles robustness in a harsh condition. The third were scaled versions of the initial sets to stress test model comparability and drive data collection. The approach targeted supervised learning as the scope focuses on DDoS attacks rather than unknown zero-day attacks, allowing the research to explore learning patterns tied to specific attack families to its full extent to contribute to robust solutions.

Eight base classifiers were ranked which used a weighted performance score that combined the metrics and the top four were chosen for the ensemble and all experimental runs. Hyperparameter tuning was set to improve performance, with the remainder kept at standard settings so the analysis focused on ensemble strategies rather than single classifiers.

3.2 Dataset generation and feature extraction

The dataset was developed using three host virtual machines, with two attackers using hping3 to implement simulated DDoS attacks simultaneously with eight attack types SYN, UDP, ICMP, ACK, SYN-ACK, RST, FIN and PSH-ACK targeting the third machine which is the victim. Benign traffic was generated separately through iperf. The eight vectors cover the main volumetric and protocol-based DDoS patterns reported in recent research and most common types used by threat actors. Each attack type was captured in a separate tcpdump pcap session. The pcap files were converted into bidirectional flow records using a custom Python Scapy based script extractor, producing a labelled dataset for supervised classification. To achieve a balanced dataset, under sampling methods were used to reduce the number of entries so that each of the nine-class types have 11,111 instances, giving a total of 99,999 flows combined in a merged dataset (per-class breakdown in Appendix A). A 9:1 attack-to-benign ratio (88,888 attack against 11,111 benign) was considered more realistic of DDoS traffic but would still introduce bias into the classifiers. All nine classes were therefore balanced equally to 11,111 flows each. Feature extraction was implemented and non-predictive and redundant fields such as flow identifiers and capture metadata were removed. Source/Destination IP addresses were also removed to prevent models from memorising attacker specific addresses, as fixed IPs from the hping3 simulation would be unrealistic in real world environments. The remaining 27 features such as packet counts, byte counts, inter-arrival times, packet sizes, rate metrics, directional ratios and TCP flag counts have characteristics directly related to DDoS such as volume and time. Hping3 was chosen for the controlled environment because the capturing process directly assigns accurate class labels, producing a fully labelled dataset for supervised classification with the conclusion addressing this limitation.

Figure 1 End-to-end construction pipeline from traffic generation to balanced dataset:



Attack Type	Description
SYN	Floods target with half-open TCP requests, exhausting connection table.
ACK	Sends large volumes of TCP ACK packets, forcing session lookups.
FIN	Sends TCP FIN packets without a valid connection, confusing stateful firewalls.
RST	Injects TCP reset packets to terminate sessions and force reconnects.
PSH-ACK	Combines PSH and ACK flags for immediate processing, targeting CPU.
SYN-ACK	Sends fake handshake replies without matching SYN, causing state mismatch.
UDP	Floods random ports with connectionless UDP, triggering ICMP unreachable replies.
ICMP	Floods target with ping requests, consuming bandwidth and CPU.

Table 1: Attack types

3.3 Classifiers and ensemble strategies

Eight supervised baseline models were selected, each chosen for their distinct learning approach and contribution to ensemble diversity. Majority of the hyperparameter's configurations were tuned for peak performances and the remaining constrained to Sklearn default standards as established baselines for reproducibility. These conditions support the stability over victory principle by removing tuning-driven variance between models.

The four selected models contribute complementary decision approaches: rule-based, linear, margin-based and neural network.

Selected base models

1. Decision Tree: A decision tree separates data into sections based on rules that allow it to model nonlinear relationships.
2. Logistic Regression: A linear baseline that uses probability to predict classes and has efficient training time.
3. SVM: Separates the data by finding the maximum margin between class types considered as effective for multi class datasets.
4. MLP: Uses neural networks to filter data through multilayered processing which allows the model to learn nonlinear patterns.

Eliminated during baseline ranking

5. Random Forest (ensemble-tree)
6. Gradient Boosting (boosting-based)
7. Naive Bayes (probabilistic)
8. KNN (instance-based)

Full hyperparameter configurations are listed on GitHub (see Reference List).

A diversity check was carried out so learning families varied with a minimum of three required. These four base models maximise disagreements to reduce predictive correlation and increase detection scope, then used to design four ensemble strategies which directly challenge each other.

The four ensemble strategy mechanisms implemented were:

1. Majority Voting: Equal voting influence across the four classifiers with the final prediction based on most votes.
2. Weighted Voting: Weights are configured manually giving priority to high performing models.
3. Stacking: All 4 base models are trained using a three-fold inner cross-validation mechanism to generate class probability outputs combined into meta-features processed by a logistic regressions meta-learning classifier for a final prediction.
4. Adaptive Voting: Dynamic weights equally assigned to each learner then adjusts after each cross-validation fold based on how each model performed (F1 macro) without manual intervention.

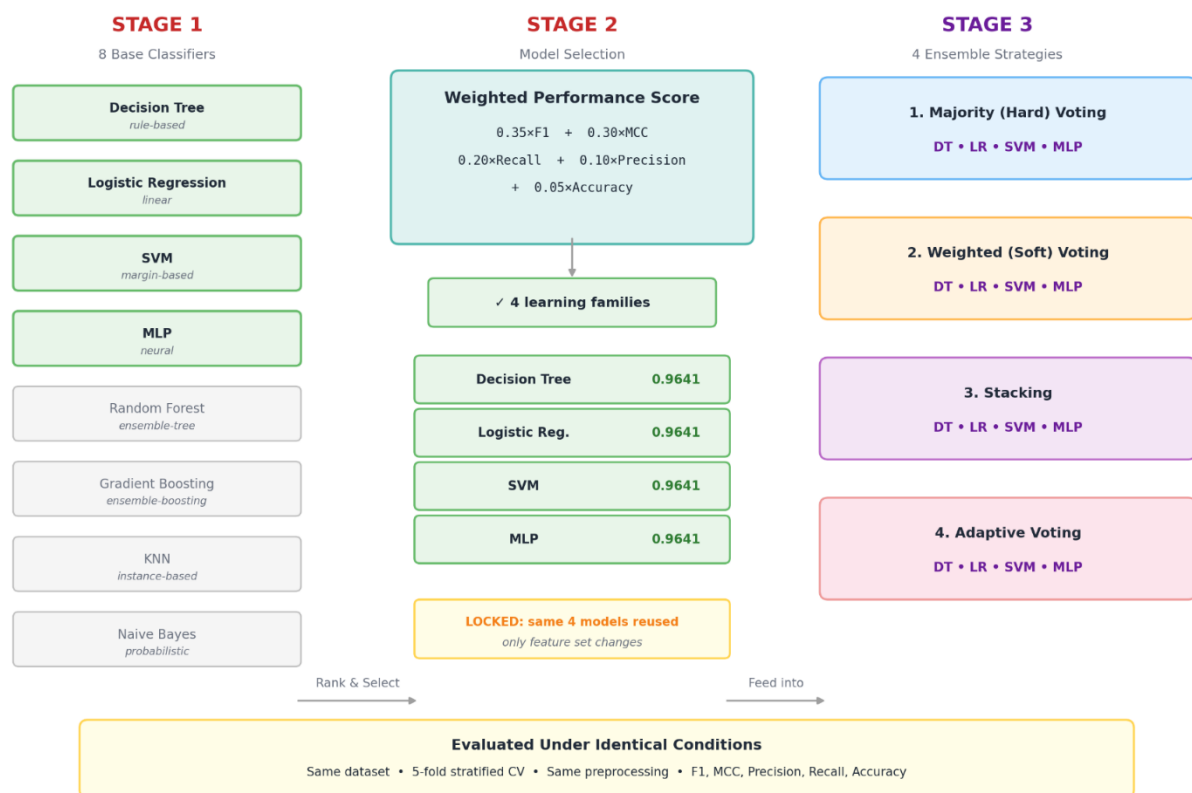


Figure 2: Three stage selection process. Baseline To Ensemble Pipeline

3.4 Experimental setup and cross-validation

The experiment used two virtualised environments. The first was VirtualBox. Three ubuntu machines were created Victim (RAM 8GB, Storage 25GB, 2 CPUs), Attacker1 and Attacker2 (RAM 4GB, Storage 16GB, 2 CPUs) to generate the nine-class type dataset within an isolated network (10.0.0.0).

The second was the Azure ML platform. To save on costs, a workspace was set in the UK South region. A Standard_D4s_v3 (4 vCPU, 16GB RAM, 32GB Storage) compute instance ran Run A and Run B. However, the compute power was upgraded for Run C and Run D due to the size increase of the dataset from 99,999 flows to 499,995, adding excessive time to runs and SVM training cost, therefore configurations continued with Standard_F16s_v2 (16 vCPU, 32GB RAM).

To measure model generalisation Five-fold stratified K-fold cross-validation was used. As a standard choice, five folds were configured, with 80% for training data and 20% for testing data in each rotation across all five iterations. In each fold, all nine classes were represented to mitigate overfitting towards a single class.

Four runs (A, B, C, D) were used against the same designed ensemble strategies. Run A used a dataset with 27 features, Run B reduced the dataset to 22 features removing (TCP flags) to evaluate how robust the model is in degraded conditions. Run C scaled Run A and Run D scaled Run B. To ensure experimental reproducibility and the continuation of strict environmental conditions, all operations used a fixed random seed of 42.

3.5 Evaluation metrics, statistics and implementation

Prioritising accuracy alone over metrics such as F1 Macro, Precision, Recall and MCC can be detrimental to multi-class evaluation, author Alharthi et al. (2025) confirming that combined multi-metric scoring is current practice in modern intrusion detection. Accuracy is 'most suited when we just care about single individuals instead of multiple classes' stated by Grandini et al. (2020), who uses a multi-class dataset. Therefore accuracy alone would be unsuitable for our nine class problem.

Accuracy is calculated as $\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$. The formula has no concept of class identity.

- Precision controls false positives by measuring how many predicted attacks are true.
- Recall measures how many actual attacks were correctly identified.
- F1 is the harmonic mean of precision and recall which can rectify any imbalances between the two.

However traditional F1 was ruled out as this is used for only single class datasets rather than representing the overall model performance across the nine-class dataset used in this study. Therefore, F1-Macro was selected as it treats each class equally, allowing equal importance towards every DDoS attack type.

MCC uses all four confusion matrix quadrants and ranges from -1 to +1. This matters in DDoS evaluation because a model labelling all traffic as benign could score high on accuracy, yet MCC would drop to 0, exposing the prediction as a random guess.

Metrics are weighted based on their importance in DDoS classification. F1 Macro ranks highest since it evaluates both precision and recall. This is especially relevant for multi-class problems with flow imbalance. Next is MCC since it reveals the models that seem to perform well but in reality, are as good as random. Recall is prioritised over precision since attacks that are not mitigated pose a long-term threat to a network. Pandey et al. (2024) demonstrates that weighted scoring metrics created an overall better performance than completely equal weight metrics. This supports our weighted formula, since the design is based on prioritised logic and not a default.

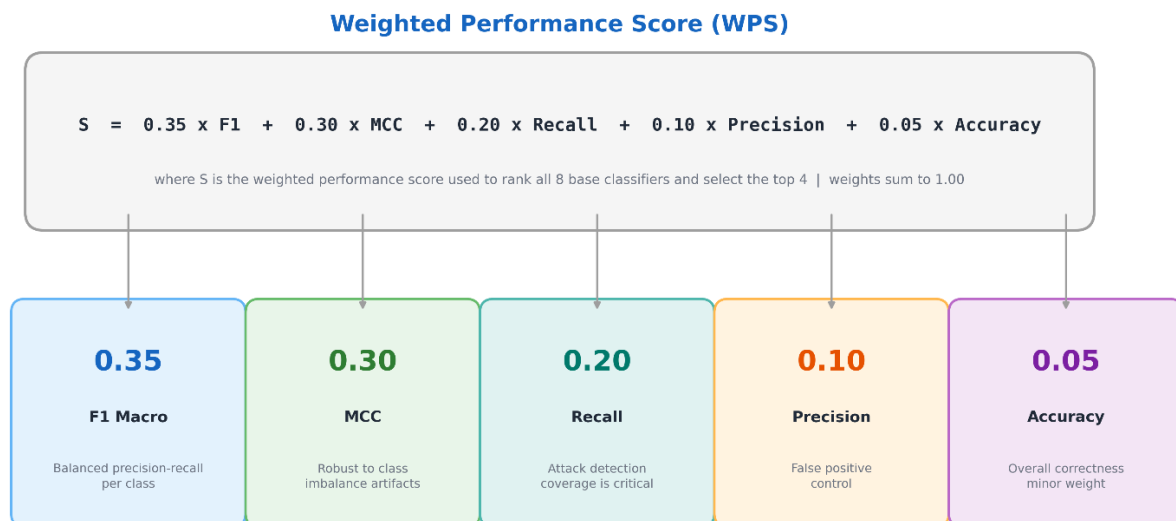


Figure 3: formula and metric weights.

3.6 Ethics

Human participants were not required for this study. Data protection and GDPR were followed and no personal data was collected. All experiments ran in isolated virtual machines using VirtualBox and Azure, with attacks contained between VMs. No external networks were affected, as vmnet and loopback addresses were used throughout. Supervisor approval was obtained for the methodology.

4. RESULTS

In this section we address the primary research question across four sub-sections. The baseline performance of the 8 classifiers, ensemble strategy comparison of the 4 Voting strategies, feature reduction impact between Run A (27 features) and Run B (22 features) and scaling impact across Run C and Run D.

4.1 Baseline classifier performance

A strong cluster was identified between the top 4 ranking models Decision Tree, Logistic Regression, SVM and MLP which all scored 0.9641 with the weighted performance formula where values are rounded to 4 decimal places and metrics such as F1 Macro, MCC, Recall, Precision and Accuracy are separated by 0.0001 per model. The bottom four Gradient Boosting, Naive Bayes, Random Forest and KNN show clear performance gaps as shown in Table 1 with weighted performance score differences of 0.0040 between Gradient Boosting and KNN and metrics separated by 0.0039 to 0.0073.

The top 4 selections were established through the weighted performance scores and a diversity check directed at the 4 distinct learning families. Rule-based, linear, margin-based and neural for the construction of robust heterogeneous ensemble strategies.

Throughout the table no single model has complete dominance. However, unexpectedly both Gradient Boosting at 5th place and Random Forest at 7th place ranked within the bottom 4 positions even though they use internal ensemble methods. Across all metrics and weighted performances scores KNN ranked the lowest. Therefore, the top four are carried forward into the ensemble mix for the comparison of ensemble strategies highlighted in Table 3.

Baseline Classifier Rankings by Weighted Performance Score

Run A | 27 Features | 99,999 Flows | 5-Fold Stratified Cross-Validation

WPS = Weighted Performance Score = 0.35×F1 + 0.30×MCC + 0.20×Recall + 0.10×Precision + 0.05×Accuracy · Values rounded to 4 d.p.

Rank	Model	F1 Macro	MCC	Recall	Precision	Accuracy	WPS	Learning Family
1	Decision Tree	0.9634	0.9619	0.9653	0.9701	0.9653	0.9641	Rule-based
2	Logistic Regression	0.9634	0.9619	0.9653	0.9701	0.9653	0.9641	Linear
3	SVM	0.9634	0.9619	0.9653	0.9701	0.9653	0.9641	Margin-based
4	MLP	0.9633	0.9619	0.9652	0.9701	0.9652	0.9641	Neural
5	Gradient Boosting	0.9633	0.9618	0.9652	0.9700	0.9652	0.9637	Ensemble-boosting
6	Naive Bayes	0.9622	0.9606	0.9641	0.9684	0.9641	0.9628	Probabilistic
7	Random Forest	0.9621	0.9604	0.9640	0.9690	0.9640	0.9626	Ensemble-tree
8	KNN	0.9560	0.9571	0.9613	0.9636	0.9613	0.9597	Instance-based

Table 2: Baseline Leaderboard

4.2 Per-class results (Ensemble)

A head-to-head comparison of the ensemble strategies results reported in Table 3 reveals that Majority Voting, Weighted Voting, Stacking and Adaptive Voting are indistinguishably tied at 4 decimal places across the metrics F1, MCC, Recall, Precision and Accuracy, with Majority Voting precisely ahead by 0.0001 at full precision value excluding the 4 d.p. ruling (see Appendix C for full 7-decimal-place rankings).

Table 2's top four models (Decision Tree, Logistic Regression, SVM and MLP) and Table 3's four strategies showed a positive correlation. The base models had extreme agreement, with average pairwise disagreement scoring 0.0% on Run A (27 feature dataset) across all 5 folds, producing identical predictions across all four strategies (see Figure 4 in 4.3).

Ensemble Strategy Comparison by F1 Macro					
Run A 27 Features 99,999 Flows 5-Fold Stratified Cross-Validation Top 4 Base Models					
Sorted by F1 Macro · Values rounded to 4 d.p.					
Strategy	F1 Macro	MCC	Recall	Precision	Accuracy
Majority Vote	0.9634	0.9619	0.9653	0.9701	0.9653
Weighted Vote	0.9634	0.9619	0.9653	0.9701	0.9653
Stacking	0.9634	0.9619	0.9653	0.9701	0.9653
Adaptive Vote	0.9634	0.9619	0.9653	0.9701	0.9653

Table 3: Ensemble Strategy Leaderboard

Panel (a) The Confusion Matrix (left side of Figure 2) represents a 9x9 grid which shows the statistics of each class (11,111 samples per class, 99,999 total). Results are identical across all 4 strategies. The heatmap uses a blue gradient representing the highest values (100%) equal to dark navy, lightest/white empty cells equal to (0%) and mid blue shades equal to the between range (icmp's 68.8% is medium navy, 19.0% syn leak is light blue, 12.2% ack leak is even lighter). However, 6 out of 9 classes achieve perfect or near perfect recall benign, psh_ack, rst, syn_ack, udp a total of (1.000) with fin achieving a near perfect classification score with only 1 sample misclassified as benign. icmp displayed a significant decline only detecting 69% of samples (7,641 out of 11,111) missing 31% (3,470 samples) which are misclassified, therefore leaking into ack and syn packets which both classify their classes correctly despite the additional icmp packets.

Panel (b) The bar chart (right side of Figure 2) displays three bars per class which represents precision, recall and F1 scores where values are perfect at 1.00 across the same 6 out of 9 classes in correlation to the confusion matrix directly reflecting the same effects. The three imperfect classes within the bar chart show the icmp leakage story previously mentioned. icmp, precision 1.00, recall 0.69, F1 0.81 (models do not incorrectly identify another class as icmp but fail to capture all). ack, precision 0.89, recall 1.00, F1 0.94 (catches every ack but 1,357 icmp samples were classified as ack). syn, precision 0.84, recall 1.00, F1 0.91 (catches every syn but 2,113 icmp samples were classified as syn)

Majority Voting led at full precision by 0.0001 but at 4 d.p. all four strategies are indistinguishable across the 99,999 dataset flows. The 0.0 disagreement indicates heterogeneity proved to be ineffective within Run A dataset size.

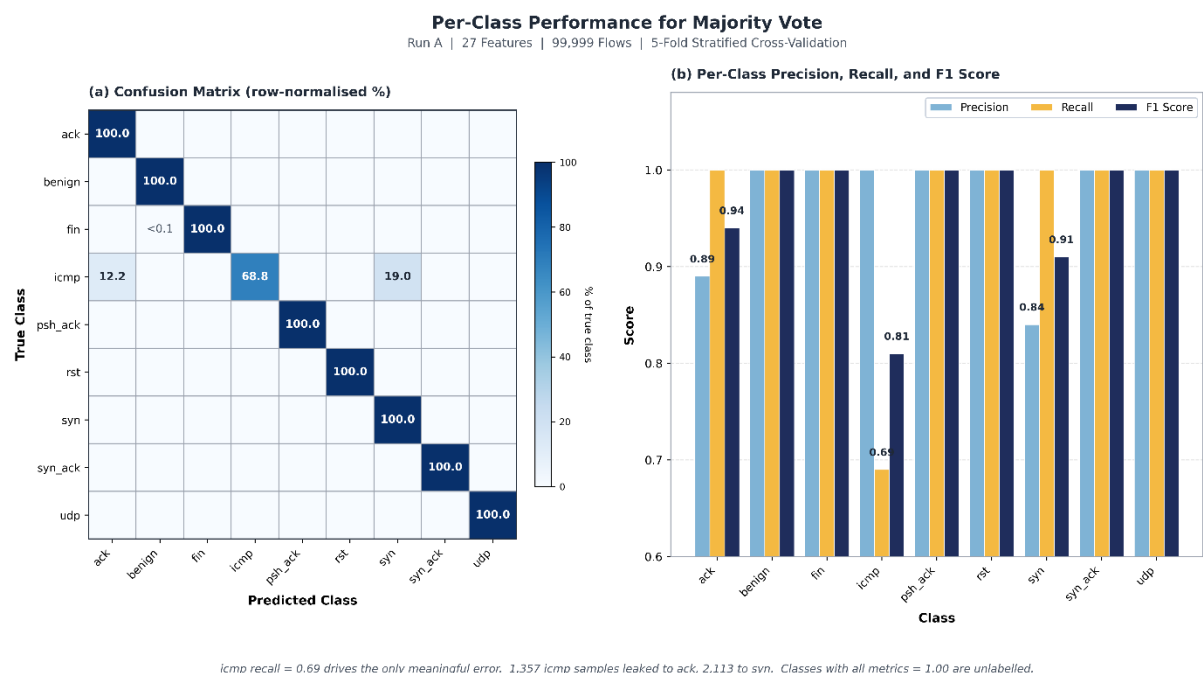


Figure 4: Per-Class Confusion Matrix and Bar-Chart Performance for Majority

4.3 Feature reduction impact (Run A vs Run B)

Table 3 shows the collapse of all four strategies. Run B's 22-feature dataset removed TCP flag count features (syn_count, ack_count, fin_count, rst_count, psh_count) from Run A's 27. Run A's leading Majority Voting F1 degrades from 0.9634 to 0.4131 in Run B and Run B leading Stacking from 0.9634 to 0.4219.

Cross-Run Strategy Delta

Run A (27 feat) vs Run B (22 feat)

Run A = 27 features | Run B = 22 features (TCP flag counts removed)

Green column = most robust strategy (smallest $|\Delta F1|$) | Red column = least robust

Macro Recall and Accuracy are mathematically equivalent under balanced multi-class labels and are reported separately for metric-completeness.

Metric	Run	Majority Vote	Weighted Vote	Stacking	Adaptive Vote
F1	Run A	0.9634	0.9634	0.9634	0.9634
	Run B	0.4131	0.4171	0.4219	0.4172
	Δ	-0.5503	-0.5463	-0.5415	-0.5462
MCC	Run A	0.9619	0.9619	0.9619	0.9619
	Run B	0.3848	0.3774	0.3823	0.3777
	Δ	-0.5771	-0.5845	-0.5796	-0.5842
Recall	Run A	0.9653	0.9653	0.9653	0.9653
	Run B	0.4386	0.4359	0.4393	0.4360
	Δ	-0.5267	-0.5294	-0.5260	-0.5293
Precision	Run A	0.9701	0.9701	0.9701	0.9701
	Run B	0.4573	0.4528	0.4599	0.4529
	Δ	-0.5128	-0.5173	-0.5102	-0.5172
Accuracy	Run A	0.9653	0.9653	0.9653	0.9653
	Run B	0.4386	0.4359	0.4393	0.4360
	Δ	-0.5267	-0.5294	-0.5260	-0.5293

Table 4: Cross-Run Strategy Delta - Run A vs Run B

Figure 4 illustrates the pairwise disagreement between Run A and Run B which moves from (0% to 39%) highlighting the previous ineffectiveness of heterogeneity appearing to be in effect when features are dropped from 27 to 22 allowing us to see the diverse learning pattern of each model's outlook for prediction. The five dots in correlation with the table below represent the scores of disagreements for each five folds during the cross-validation tuning and training process where the mean score is then selected as the primary value. Run A showed an almost constant agreement throughout the fold with scores ranging from 0.00% to 0.01% and Run B's disagreement ranged from 34.6% to 41.6%. Stacking led Run B with $F1=0.4219$ and Majority Voting followed with $F1=0.4131$ across the 39% disagreement environment.

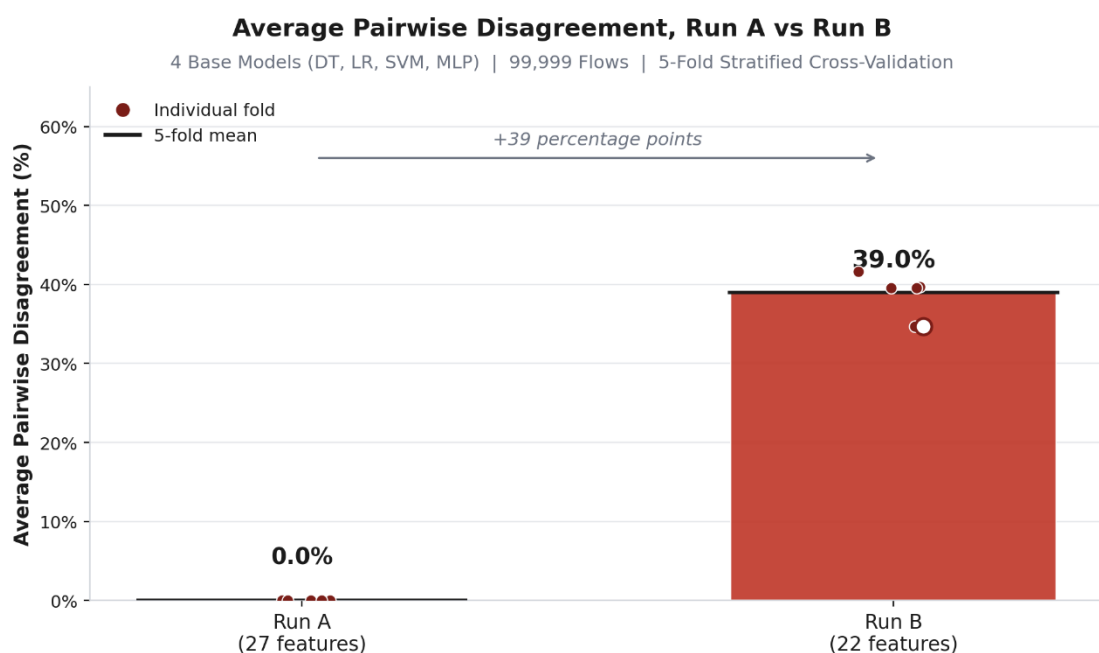


Figure 5: Pairwise disagreement Bar-Chart Run A vs Run B

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean
Run A (27 feat)	0.00%	0.00%	0.01%	0.01%	0.01%	0.01%
Run B (22 feat)	39.72%	41.65%	39.52%	34.65%	39.53%	39.01%

Table 5: Average Pairwise Disagreement Per-Fold, Run A vs Run B

Dots show per-fold disagreement; horizontal bar is the 5-fold mean. Pairwise disagreement measures the fraction of predictions on which the four base models disagree.

Panel (a) The Confusion Matrix (left side of Figure 3) represents a 9x9 grid which shows the statistics of each class (11,111 samples per class, 99,999 total). Results reflect Stacking, the leading strategy within Run B. icmp's 68.8% is medium navy, syn_ack's 42.7%, psh_ack's 41.5%, ack's 35.5% syn_ack leak are lighter shades. However, only 2 out of 9 classes achieve perfect recall benign, udp a total of (1.000) with icmp retaining partial classification at 68.8% despite the feature reduction. ack, fin, psh_ack, rst, syn and syn_ack displayed a significant decline collapsing into the flag-family cluster, therefore leaking into each other as flag count features were removed which previously defined their separation.

Panel (b) The bar chart (right side of Figure 3) shows precision, recall and F1 perfect at 1.00 for the same 2 out of 9 classes from the confusion matrix. icmp, precision 1.00, recall 0.69, F1 0.81 (models do not incorrectly identify another class as icmp but fail to capture all) and the remaining six flag classes (ack, syn, rst, fin, psh_ack, syn_ack) all collapsed with F1 scores between 0.02 and 0.27, confirming the feature-label coupling between flag classes and flag count features.

Run B's 22 feature dataset reduction revealed that Stacking is a leading strategy which thrives within heterogeneity and disagreement environments (0% to 39%) but also suffers collapse ($\Delta F1 \approx -0.55$). All the ensemble strategies are still closely performing, this prompted further inquiry into the scalability of the dataset size within Run C and Run D.

Per-Class Performance for Stacking

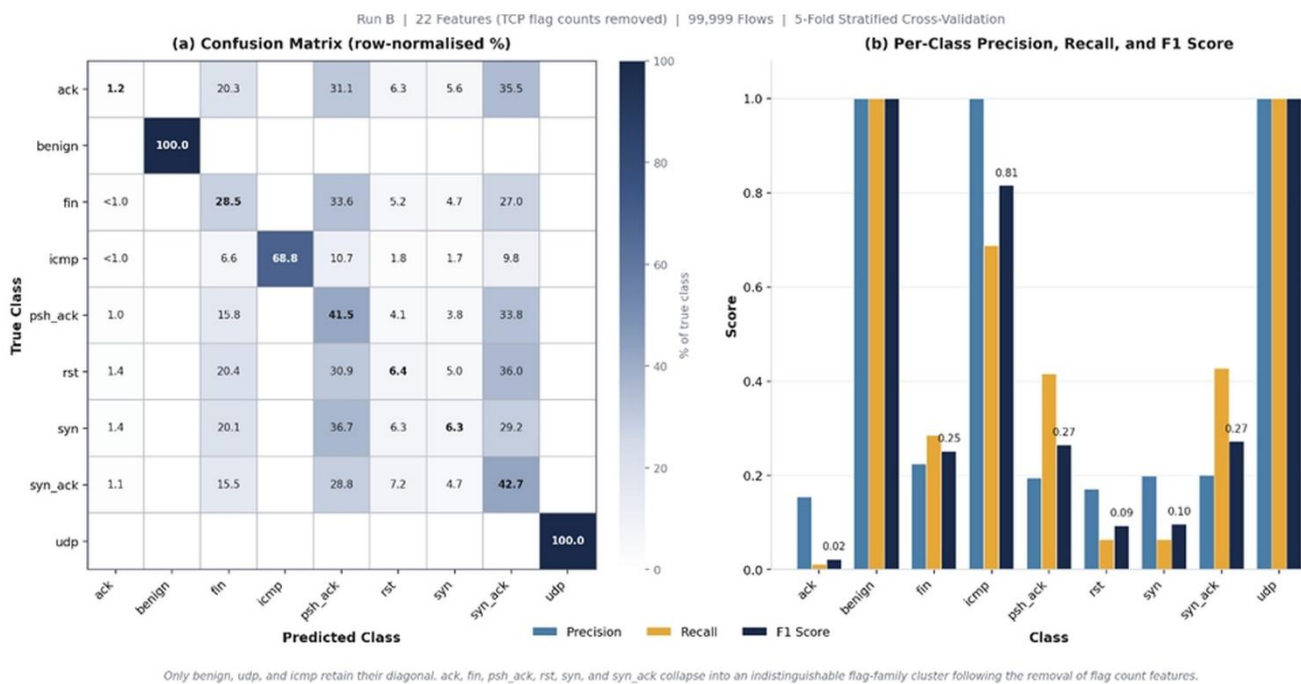


Figure 6: Per-Class Confusion Matrix and Bar-chart

4.4 Scaling impact (Run C and Run D)

Table 4 introduces the results of the scaled versions of A and B across the five main metrics F1 Macro, MCC, Recall, Precision and Accuracy where scaling started from a total dataset size jump from 99,999 flows in Runs A and B to 499,995 flows in Runs C and D. Run A scores all tie at 4dp for all strategies. F1 (0.9634), MCC (0.9619), Recall (0.9653), Precision (0.9701) and Accuracy (0.9653).

The rest of the metrics identify ranges rather than equality. Run B F1 scores range from 0.4131 to 0.4219, MCC starts from 0.3774 to 0.3848. Recall scores are 0.4359 to 0.4393, Precision at 0.4528 to 0.4599 and Accuracy at 0.4359 to 0.4393. Run C, F1 scores have a difference of 0.0001 between strategies, MCC starts from 0.9616 across all strategies. Recall scores are 0.9649 to 0.9650, Precision at 0.9698 to 0.9699 and Accuracy at 0.9649 to 0.9650. Run D, F1 scores range from 0.4179 (Majority Vote) to 0.4230 (Stacking), MCC starts from 0.3817 to 0.3859. Recall scores are 0.4391 to 0.4422, Precision at 0.4479 to 0.4685 and Accuracy at 0.4391 to 0.4422.

The deltas across runs identify the scaling effect at fixed feature count. $\Delta F1$ (A $\rightarrow$ C) ranges from -0.0003 to -0.0004 across all four strategies, MCC starts from -0.0003 across all strategies. Recall and Accuracy at -0.0003 to -0.0004 and Precision at -0.0002 to -0.0003. $\Delta F1$ (B $\rightarrow$ D) ranges from +0.0011 (Stacking) to +0.0048 (Majority Vote), MCC starts from -0.0003 (Majority Vote) to +0.0043 (Weighted Vote). Recall and Accuracy at +0.0019 to +0.0032 and Precision at -0.0094 (Majority Vote) to +0.0086 (Stacking).

F1 standard deviation across folds in Run D ranges from 0.000971 (Weighted Vote) to 0.001799 (Majority Vote), with Stacking at 0.001400. In Run C the std ranges from 0.000235 (Stacking) to 0.000307 across all other strategies. (full per-fold metrics for all four runs in Appendix B).

Cross-Run Strategy Comparison: All Four Experimental Runs

F1 Macro						
Strategy	F1 Run A	F1 Run B	F1 Run C	F1 Run D	$\Delta F1$ (A $\rightarrow$ C)	$\Delta F1$ (B $\rightarrow$ D)
Majority Vote	0.9634	0.4131	0.9631	0.4179	-0.0003	+0.0048
Weighted Vote	0.9634	0.4171	0.9631	0.4189	-0.0003	+0.0018
Stacking	0.9634	0.4219	0.9630	0.4230	-0.0004	+0.0011
Adaptive Vote	0.9634	0.4172	0.9631	0.4189	-0.0003	+0.0017
Matthews Correlation Coefficient (MCC)						
Strategy	MCC Run A	MCC Run B	MCC Run C	MCC Run D	ΔMCC (A $\rightarrow$ C)	ΔMCC (B $\rightarrow$ D)
Majority Vote	0.9619	0.3848	0.9616	0.3845	-0.0003	-0.0003
Weighted Vote	0.9619	0.3774	0.9616	0.3817	-0.0003	+0.0043
Stacking	0.9619	0.3823	0.9616	0.3859	-0.0003	+0.0036
Adaptive Vote	0.9619	0.3777	0.9616	0.3817	-0.0003	+0.0040
Recall (Macro)						
Strategy	Rec Run A	Rec Run B	Rec Run C	Rec Run D	ΔRec (A $\rightarrow$ C)	ΔRec (B $\rightarrow$ D)
Majority Vote	0.9653	0.4386	0.9650	0.4405	-0.0003	+0.0019
Weighted Vote	0.9653	0.4359	0.9650	0.4391	-0.0003	+0.0032
Stacking	0.9653	0.4393	0.9649	0.4422	-0.0004	+0.0029
Adaptive Vote	0.9653	0.4360	0.9650	0.4392	-0.0003	+0.0032
Precision (Macro)						
Strategy	Prec Run A	Prec Run B	Prec Run C	Prec Run D	$\Delta Prec$ (A $\rightarrow$ C)	$\Delta Prec$ (B $\rightarrow$ D)
Majority Vote	0.9701	0.4573	0.9699	0.4479	-0.0002	-0.0094
Weighted Vote	0.9701	0.4528	0.9699	0.4583	-0.0002	+0.0055
Stacking	0.9701	0.4599	0.9698	0.4685	-0.0003	+0.0086
Adaptive Vote	0.9701	0.4529	0.9699	0.4583	-0.0002	+0.0054
Accuracy						
Strategy	Acc Run A	Acc Run B	Acc Run C	Acc Run D	ΔAcc (A $\rightarrow$ C)	ΔAcc (B $\rightarrow$ D)
Majority Vote	0.9653	0.4386	0.9650	0.4405	-0.0003	+0.0019
Weighted Vote	0.9653	0.4359	0.9650	0.4391	-0.0003	+0.0032
Stacking	0.9653	0.4393	0.9649	0.4422	-0.0004	+0.0029
Adaptive Vote	0.9653	0.4360	0.9650	0.4392	-0.0003	+0.0032

Run A = 27 features, 99,999 flows | Run B = 22 features, 99,999 flows | Run C = 27 features, 499,995 flows | Run D = 22 features, 499,995 flows
Green = most robust strategy (highest values in heterogeneity-rich runs). Red = least robust strategy. Δ columns show scaling effect at fixed feature count.
Macro Recall and Accuracy are mathematically equivalent under balanced multi-class labels and are reported separately for metric-completeness.

Table 6: Cross-Run Ensemble Comparison All metrics, All Four runs

5. DISCUSSION

In this section we revisit the results to analyse the four runs and interpret the effects of heterogeneity, scaling and feature reduction through an in-depth examination. Majority Voting, Weighted Voting, Adaptive Voting and Stacking are compared head-to-head under identical conditions to reach stability over victory.

5.1 Feature-reduction tradeoff and heterogeneity collapse

The clustered performance spread across the top 4 models makes them an effective contribution towards the ensemble. No one model regulates another or dominates the team, allowing equal contribution to predictions. Considering Run A (27 feature dataset) and Run B (22 feature dataset) model performance is at its peak during the initial run however declines in efficiency when feature reduction has taken place. Despite different learning families, heterogeneity did not take place as the four strategies produced identical predictions. The hypothesis was confirmed within this research, no ensemble strategy had complete dominance, however the mathematical tie where all four strategies performed equally at 4dp revealed a heterogeneity collapse, shifting the tradeoff to stability over victory. This supports Hancock et al. (2023) who argues that accuracy alone can be misleading, where Run B's accuracy of 43.86% looks recoverable on its own but an F1 score of 0.4131 and MCC score show the full drop in performance, which is why the weighted performance score was used to down-weight accuracy.

Stacking leads within Run B because it uses the meta-learning abilities which is functioned to specifically resolve disagreements through pattern learning and frontline predictions before the outcome of the prioritised model. However Majority Voting fails in this case due to model disagreement as it goes against its core functions such as majority consensus, if this requirement is not met the outcome leads to a default tiebreak which leads to a random decision. This supports Dietterich (2000) who established that ensemble diversity depends on the feature space, where Run B's identical collapse across all four strategies shows that reducing features from 27 to 22 removed the diversity that Voting and Stacking strategies need to perform. Adaptive Voting performed evenly with the static voters in Run A and shared the same collapse in Run B as the work of Mulimani and Patil's (2026) mentioned the flaws of Adaptive Voting and the benefit of the ensemble having full efficiency on live traffic and not fixed datasets.

5.2 Scale-only validation and stability over victory (Run C)

The evaluation between Run A and Run C is vital as it is a direct development of one another to assess a scale only test with the 27 features. The comparison was designed to expose whether scale alone drives change across three elements. Heterogeneity which questions whether the scale allows the models to make diverse decisions. Pairwise disagreement questions whether the scale forces disagreement between models. Strategy margins questions whether scaling separates F1 scores between strategies. The top F1 scores for all models with 27 features were led by Majority Voting by a small metric separation rounded to 4dp with the remaining models following closely. All ensemble strategies on Run A returned F1=0.9634, MCC=0.9619 and Accuracy=0.9653. However, on Run C all Voting strategies returned F1=0.9631, MCC=0.9616 and Accuracy=0.9650 with Stacking trailing by 0.0001 at F1=0.9630. The diminishing differences between Run A and C range from 0.0003 to 0.0004 which has made very minor movement slightly decreasing from scaling but nothing significant to affect the ranked positions preserved due to scaling not improving each model's peak performance. The heterogeneity structure collapsed because the weights of the base models became equal, therefore diversity is lowered when constructed into ensemble strategies as the base models are already outputting closely on their own. This is shown by Majority Voting and Weighted Voting returning identical F1 scores within 7 decimal places (0.9630541540 vs 0.9630541538). Pairwise agreement values display slight disagreement through training folds with values displaying that the disagreement varies across folds. Average disagreement was 0.0126% with fold 1 spiking to 0.0525% while folds 2-5 averaged 0.0026%. Strategy margins highlight that Stacking F1 standard deviation is 0.000235 compared to 0.000307 for the three voters meaning Stacking varied less across the 5 folds making it the most stable strategy at scale.

The finding of this study provides direct contributions towards the research question 'Which heterogeneous ensemble strategy is most effective for flow-based multi-class DDoS detection?' The independent variable of scale could not be the deciding factor as it could not separate strategies based on core metrics as ranking held, so no dominant strategy could be declared superior. Stacking trades 0.0001 mean F1 for 24% lower fold variance. Voting strategies converge on performance while Stacking stands apart on stability.

5.3 Statistical confirmation via McNemar test

Figure 5 shows the McNemar's Test contingency tables for each of the ensemble strategies, comparing Run A (F1=0.9634) against Run B (F1=0.4131). All four strategies produced $\chi^2 > 52,000$, $p < 0.001$, meaning the difference between Run A and Run B was statistically significant. The structures are four square grids where the cells are the predictions for Run A and Run B. The first line shows Majority Voting, where it predicted 43,856 incorrect for both, predicting 52,672 incorrect on Run A, summing 96,528 out of 99,999 flows predicted correctly for the accuracy score of 96.53% shown in Table 2. The bottom left (orange cell) shows that Majority Voting had 0 cases where Run A was incorrect and Run B was correct. The bottom right (red cell) shows 3,471 incorrect on both runs, which is why Run A has a total of 96,528 with the remaining 3,471 making up full coverage of the 99,999 flows. The bottom left cells across all four strategies show counts of 0, 1, 1, 1, meaning Run B's correct predictions were already covered by Run A and the 22-feature dataset added no unique correctness anywhere in the test set.

Across all four strategies the collapse pattern was the same, with χ^2 values clustered between 52,594 and 52,930, showing that the 22-feature reduction damaged every ensemble equally regardless of architecture and that no strategy could compensate for the loss of the five removed features. Where Oyucu et al. (2023) reports ensemble accuracy above 90% on SDN-SCADA hping3 dataset as evidence of robustness, Run A confirms this with F1=0.9634 but Run B exposes a feature-reduction tradeoff Oyucu et al. does not address, where ensemble robustness collapses across all condition including when the feature space is degraded.

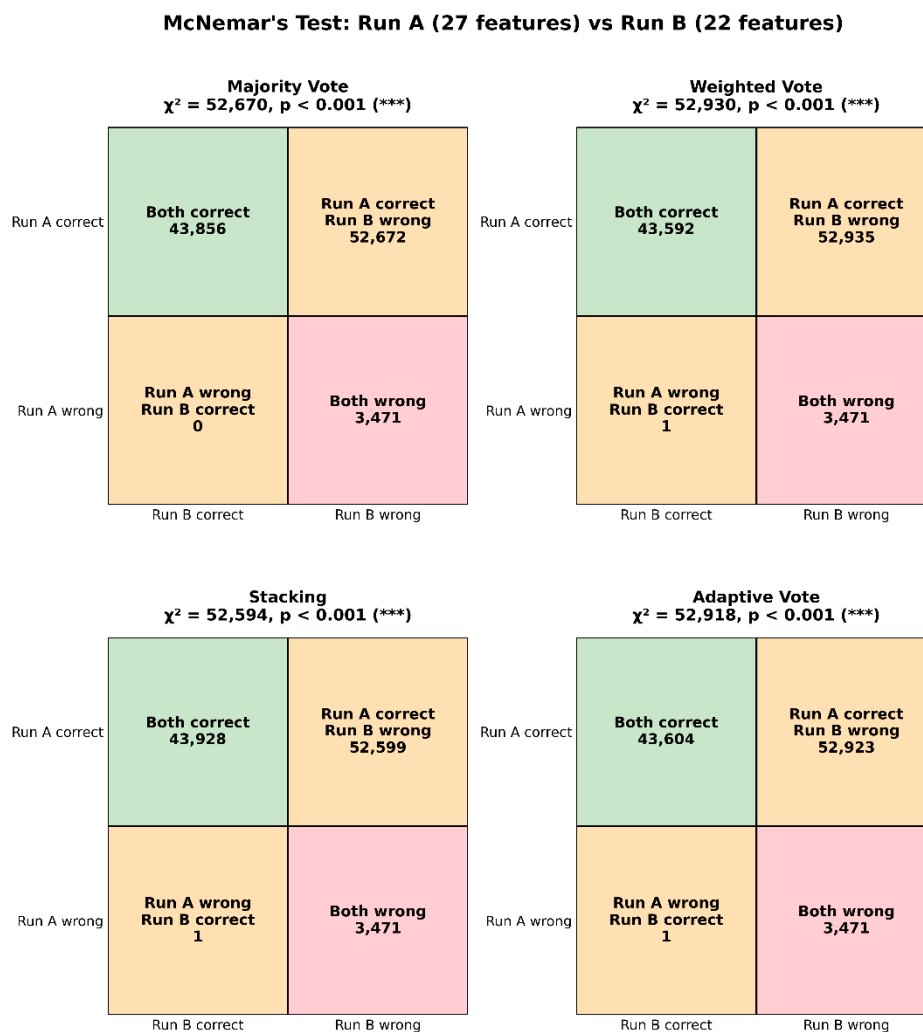


Figure 7: McNemar's Test

5.4 Production-scale validation (Run B vs Run D)

Scaled runs revealed the relationship between disagreement, heterogeneity and feature reduction. The reduced features drive heterogeneity, demonstrating that baseline classifiers behave differently to non-linear traffic and heterogeneity matters for insight that can strengthen controls and improve IDS performance. Disagreement grouped by feature set, not by dataset size, the four base classifiers split closely on the 22-feature runs and stayed aligned on the 27-feature runs (see Appendix A for the full feature list). One interpretation is that the 27-feature dataset had features (specifically, TCP flags) that were so informative in relation to the traffic from hping3 that all four base classifiers agreed on the same decision boundary, concealing their true distinctiveness. When those features were absent in Run B and Run D, the classifiers had to work beyond the most predictive shortcuts, uncovering the true disagreement.

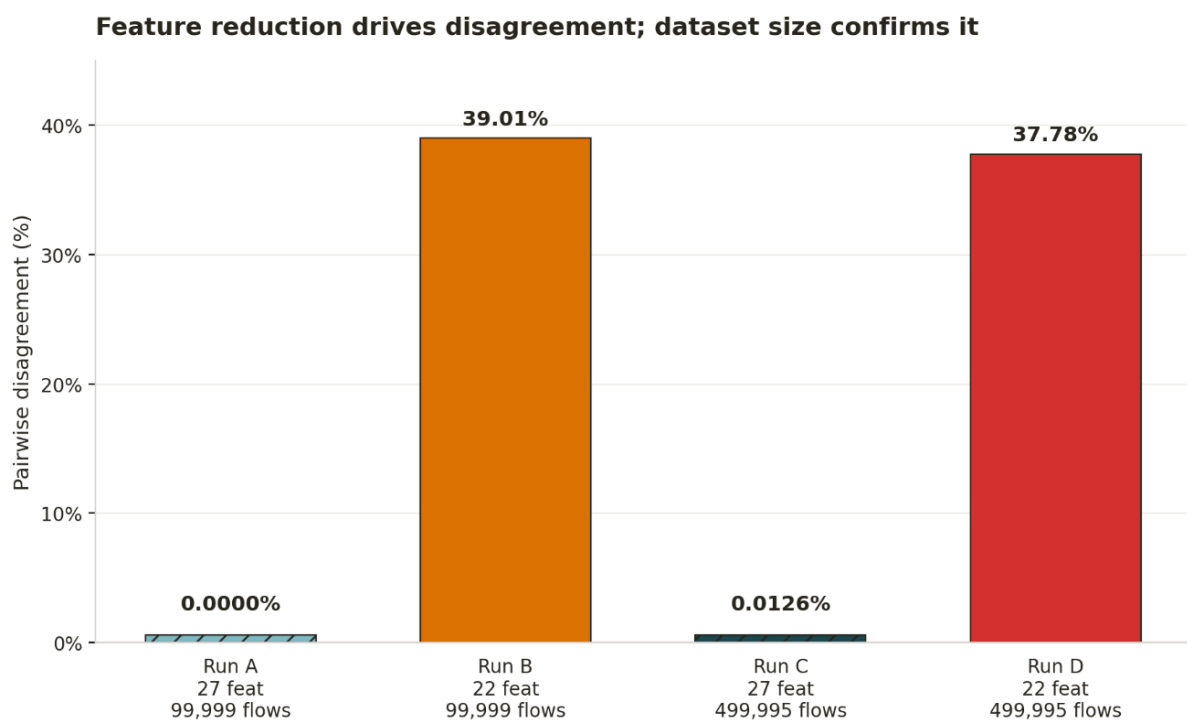


Figure 8: what drives disagreement? - Overview Bar-chart of all runs

6. CONCLUSIONS

We addressed the gap of limited head-to-head ensemble strategy comparison under fair conditions. The research answered the question, 'Which heterogeneous ensemble strategy is most effective for flow based multi class DDoS detection?' Production DDoS systems need real-world reliability and generalisation capability to non-linear traffic where defenders rely on stability over victory.

6.1 Findings

The first finding came from Run A. Testing all four ensemble strategies revealed a mathematical tie where all four strategies performed equally at 4dp, revealing the heterogeneity collapse as the models' predictions were identical and 0.0% disagreement on Run A. Majority Voting edged ahead at 7dp on metric averaging, due to the floating-point differences, not prediction differences.

Following this, the second finding came from Run B. Raising heterogeneity across models by reducing the features (TCP flag removal) responded to the ineffective contribution within Run A's test. This forced disagreement through feature reduction, raising pairwise disagreement from 0% in Run A to 39% in Run B which allowed us to analyse the results from a different perspective. However, all strategies degraded ($\Delta F1 \approx -0.55$) but at different rates allowing Stacking to lead under disagreement conditions due to its meta-learning abilities.

The third finding came from Run C. Stress testing the ensemble strategies by increasing the volume of flows tested the robustness of models under heavier workloads. However, we discovered that scaling did not change the rankings, there were no dominant performers, but did reveal deeper precision with Stacking F1 standard deviation 0.000235 vs voters 0.000307 (24% less variance).

The fourth finding was from Run D, which scored a disagreement percentage of 37.78%, similar to Run B's 39.01%. Scaling the dataset to 499,995 flows resulted in a difference of only 1.23% compared to Run B. This confirmed that scale was not responsible for the disagreement results which were caused by feature reduction. Therefore, Stacking led, showing versatility throughout runs when F1 margins are close.

Across all four runs, the answer to the research question is that no single strategy dominated on F1 alone, but Stacking emerged as the most effective strategy on the basis of stability over victory.

6.2 Recommendations

The first recommendation is at the design level. Feature engineering should be prioritised over dataset volume. The second recommendation is using a customised weighted performance score over single metric ranking to tradeoff on metrics so that ensemble strategies may provide more effectiveness based on use case needs at the deployment level. The third recommendation is stacking for production DDoS detection where stability matters most or Voting strategies when compute cost is the preferred priority.

6.3 Limitations

This study was limited by the controlled testing environment, which may reduce the generalisation of results to more diverse traffic patterns. Therefore, the three core areas which provided limitations were methodology constraints, scope constraints and resource constraints.

Methodology constraints involved manual hyperparameter tuning as limited time did not allow configurations to be exhaustively searched, meaning the closely performing ensemble strategies may have been more separable if automated tuning had been used. Testing baseline ranking on Run B revealed a limitation. The top four classifiers from Run A, which were fixed for the rest of the runs, were no longer the top four under the reduced feature set. Logistic Regression became less effective within the ensemble.

Scope was limited in areas regarding the wider variety of design choices for ensemble strategies such as deep learning models. Internal and external ensemble strategies were considered for base classifiers but were excluded to preserve the fair condition, which limited the extent of direct comparisons to the broader ensemble research. The second scope related limitation is the static dataset experiments with known attacks, which excluded streaming traffic evaluation and zero-day attacks.

Resource constraints included consumer-grade hardware as GPU can be much more efficient when training models where folds can be increased without affecting the time training lasts and SVM which has extremely slow training times affecting the final Run D.

6.4 Future Work

Methodology directions. The first element that the future work may also consider revisiting base classifier selection for each feature condition, but without constraining the ensemble composition from Run A, to determine whether ensemble disagreement occurs as a result of only feature reduction or due to the retention of suboptimal classifiers.

Scope directions. The second area future work may consider is extending the scope beyond heterogeneous ensemble strategies under controlled conditions to explore strategy extensions such as homogeneous methods and hybrid approaches including (ML/DL) ensembles. Validation could also be extended through common research datasets such as CICDDoS2019 for benchmark comparison, zero-day evaluation for the validation of reliability and streaming traffic to fully test the ability of dynamic traffic with Adaptive Voting.

Resource directions. The third part that should be reviewed is the compute power of the device as enterprise or scientific level hardware can be the solution for the two specific bottlenecks this study encountered. The first encounter was the SVM training stall which could be eliminated. And the second encounter is the Fold counts in cross-validation could increase without lengthening training time.

Together, these directions form a proactive step toward more reliable systems where stability over victory becomes a deployable principle for production environments.

6.5 Contributions

This section closes the three literature gaps with three solutions. The first gap, identified by Maodah et al. (2025), is answered by the head-to-head comparison of Voting and Stacking ensemble strategies across four runs. The second gap, confirmed by the studies of Cheng and Feng (2026), is answered by the stability assessment through Runs A, B, C and D. The third gap, identified by Lucas et al. (2023), is answered by the solution to this comparative study, which stands as the stability over victory approach, which gives practitioners guidance for reproducibility, developments and ensemble strategy selection in production environments. This research created a focused contribution by addressing identified gaps in a single experiment.

ACKNOWLEDGEMENTS

After years of late-night studying while working, I have reached this final piece of written work, which has been an insightful process to remember. Acknowledging every interaction that has contributed towards my educational success and personal growth at Sheffield Hallam University, because without those interactions my journey would not have been the same. And a big thank you to John Heggarty for his guidance and support that helped me reach my research goals.

REFERENCES

- Abedin, T., Xu, H., & Uddin, S. (2026). The impact of K selection in K-fold cross-validation on bias and variance in supervised learning models. *Scientific Reports*, 16(1), 6084.
<https://doi.org/10.1038/s41598-026-37247-x>
- Acharya, T., Khatri, I., Annamalai, A., & Chouikha, M. F. (2021). Efficacy of Heterogeneous Ensemble Assisted Machine Learning Model for Binary and Multi-Class Network Intrusion Detection. *2021 IEEE International Conference on Automatic Control & Intelligent Systems (I2CACIS)*, 408–413.
<https://doi.org/10.1109/I2CACIS52118.2021.9495864>
- Alashhab, A. A., Zahid, M. S., Isyaku, B., Elnour, A. A., Nagmeldin, W., Abdelmaboud, A., Abdullah, T. A. A., & Maiwada, U. D. (2024). Enhancing DDoS Attack Detection and Mitigation in SDN Using an Ensemble Online Machine Learning Model. *IEEE Access*, 12, 51630–51649. <https://doi.org/10.1109/ACCESS.2024.3384398>
- Alharthi, M., Medjek, F., & Djenouri, D. (2025). Ensemble Learning Approaches for Multi-Class Intrusion Detection Systems for the Internet of Vehicles (IoV): A Comprehensive Survey. *Future Internet*, 17(7), 317. <https://doi.org/10.3390/fi17070317>
- Angulo, E., Lizcano, L., & Marquez, J. (2026). A Stacking-Based Ensemble Model for Multiclass DDoS Detection Using Shallow and Deep Machine Learning Algorithms. *Applied Sciences*, 16(2), 578. <https://doi.org/10.3390/app16020578>
- Cheng, S., & Feng, X. (2026). Ensemble-based detection of distributed denial-of-service attacks in IoT networks using majority decision mechanisms. *Scientific Reports*, 16(1), 10886.
<https://doi.org/10.1038/s41598-025-23638-z>
- Cloudflare. (n.d.). *Famous DDoS attacks: The largest DDoS attacks of all time*. Cloudflare Learning Center. <https://www.cloudflare.com/en-gb/learning/ddos/famous-ddos-attacks/>

- Cloudflare. (2026a, February 5). *DDoS threat report for 2025 Q4*. The Cloudflare Blog.
<https://blog.cloudflare.com/ddos-threat-report-2025-q4/>
- Cloudflare. (2026b, March 3). *Cloudflare 2026 Threat Report*. The Cloudflare Blog.
<https://blog.cloudflare.com/2026-threat-report/>
- Dietterich, T. G. (2000). Ensemble methods in machine learning. In J. Kittler & F. Roli (Eds.), *Multiple Classifier Systems* (Vol. 1857, pp. 1–15). Springer. https://doi.org/10.1007/3-540-45014-9_1
- Grandini, M., Bagli, E., & Visani, G. (2020). Metrics for Multi-Class Classification: An Overview. *ArXiv*. <https://doi.org/10.48550/arXiv.2008.05756>
- Hancock, J. T., Khoshgoftaar, T. M., & Johnson, J. M. (2023). Evaluating classifier performance with highly imbalanced Big Data. *Journal of Big Data*, 10(1), 42.
<https://doi.org/10.1186/s40537-023-00724-5>
- Hasanah, M., Putri, R. A., Aidie, M., Putra, R., & Ahmad, T. (2024). Analysis of Weight-Based Voting Classifier for Intrusion Detection System. *International Journal of Intelligent Engineering and Systems*, 17(2), 190–200. <https://doi.org/10.22266/ijies2024.0430.17>
- Jiyad, Z. M., Maruf, A. Al, Haque, Md. M., Gupta, M. Sen, Ahad, A., & Aung, Z. (2024). DDoS Attack Classification Leveraging Data Balancing and Hyperparameter Tuning Approach Using Ensemble Machine Learning with XAI. *2024 Third International Conference on Power, Control and Computing Technologies (ICPC2T)*, 569–575.
<https://doi.org/10.1109/ICPC2T60072.2024.10475035>
- Kim, H., Ham, D., & Moon, K.-S. (2025). Adaptive Sampling Framework for Imbalanced DDoS Traffic Classification. *Sensors*, 25(13), 3932. <https://doi.org/10.3390/s25133932>
- Lucas, T. J., de Figueiredo, I. S., Tojeiro, C. A. C., de Almeida, A. M. G., Scherer, R., Brega, J. R. F., Papa, J. P., & da Costa, K. A. P. (2023). A Comprehensive Survey on Ensemble Learning-Based Intrusion Detection Approaches in Computer Networks. *IEEE Access*, 11, 122638–122676. <https://doi.org/10.1109/ACCESS.2023.3328535>

- Maodah, K. A., Alhomdy, S., & Thabit, F. (2025). Detecting intrusions in cloud-based ensembles: evaluating voting and stacking methods with machine learning classifiers. *Frontiers in Computer Science*, 7. <https://doi.org/10.3389/fcomp.2025.1623375>
- Mulimani, D. C., & Patil, P. R. (2026). An Adaptive Machine Learning Model for Heterogeneous Concept Drift Handling in Streaming Data. *Engineering, Technology & Applied Science Research*, 16(2), 33672–33677. <https://doi.org/10.48084/etasr.17430>
- Mvalo, S. (2026). *Research Methods [Lecture notes]*. Sheffield Hallam University.
- Oyucu, S., Polat, O., Türkoğlu, M., Polat, H., Aksöz, A., & Ağdaş, M. T. (2023). Ensemble Learning Framework for DDoS Detection in SDN-Based SCADA Systems. *Sensors*, 24(1), 155. <https://doi.org/10.3390/s24010155>
- Pandey, G., Bagri, R., Gupta, R., Rajpal, A., Agarwal, M., & Kumar, N. (2024). Robust weighted general performance score for various classification scenarios. *Intelligent Decision Technologies*, 18(3), 2033–2054. <https://doi.org/10.3233/IDT-240465>
- Saunders, M., Lewis, P., & Thornhill, A. (2019). *Research methods for business students* (8th ed.). Pearson Education.
- Verma, P., Kowsik, A. R. K., Pateriya, R. K., Bharot, N., Vidyarthi, A., & Gupta, D. (2024). A Stacked Ensemble Approach to Generalize the Classifier Prediction for the Detection of DDoS Attack in Cloud Network. *Mobile Networks and Applications*, 29(5), 1618–1632. <https://doi.org/10.1007/s11036-023-02225-4>

APPENDICES

Appendix A. Full per-class table

Table 7: Appendix A - Full Feature List

Feature Set Comparison: Run A (27) vs Run B (22)				
Five TCP flag count features removed in Run B to force ensemble disagreement				
#	Feature Name	Category	Run A (27)	Run B (22)
1	src_port	Identifier	✓	✓
2	dst_port	Identifier	✓	✓
3	protocol	Identifier	✓	✓
4	flow_duration	Temporal	✓	✓
5	pkt_count_fwd	Packet count	✓	✓
6	pkt_count_bwd	Packet count	✓	✓
7	pkt_count_total	Packet count	✓	✓
8	byte_count_fwd	Byte count	✓	✓
9	byte_count_bwd	Byte count	✓	✓
10	byte_count_total	Byte count	✓	✓
11	mean_iat	Inter-arrival time	✓	✓
12	min_iat	Inter-arrival time	✓	✓
13	max_iat	Inter-arrival time	✓	✓
14	std_iat	Inter-arrival time	✓	✓
15	mean_pkt_size	Packet size	✓	✓
16	min_pkt_size	Packet size	✓	✓
17	max_pkt_size	Packet size	✓	✓
18	bytes_per_pkt	Rate	✓	✓
19	pkt_rate	Rate	✓	✓
20	byte_rate	Rate	✓	✓
21	fwd_bwd_pkt_ratio	Directional ratio	✓	✓
22	fwd_bwd_byte_ratio	Directional ratio	✓	✓
23	syn_count	TCP flag	✓	✗
24	ack_count	TCP flag	✓	✗
25	fin_count	TCP flag	✓	✗
26	rst_count	TCP flag	✓	✗
27	psh_count	TCP flag	✓	✗

Figure 9: Full Feature Description List

Legend		
✓	Feature included	✗ Feature removed in Run B
	Run A and Run C use the full 27 features. Run B and Run D use the reduced 22 features.	
		■ Dropped TCP flag features
Feature Category Descriptions		
Brief explanation of what each feature category captures from the flow.		
Identifier		
What it is:	Source port number, destination port number, and the transport-layer protocol used by the flow (TCP, UDP, or ICMP).	
DDoS relevance:	Separates UDP and ICMP floods from TCP-based attacks and exposes port-targeting behaviour where attackers concentrate traffic on specific service ports.	
Temporal		
What it is:	Total duration of the flow in seconds, measured from the first packet timestamp to the last.	
DDoS relevance:	DDoS flows are typically short and bursty whereas benign flows tend to last longer with sustained interaction.	
Packet count		
What it is:	Number of packets sent in the forward direction, the backward direction, and the combined total across the flow.	
DDoS relevance:	Floods produce extremely high forward packet counts with little or no backward response which signals one-way attack traffic.	
Byte count		
What it is:	Total bytes transferred in the forward direction, the backward direction, and the combined total across the flow.	
DDoS relevance:	Volumetric attacks show heavy one-directional byte transfer with no return traffic which indicates a target being overwhelmed rather than a normal exchange.	
Inter-arrival time		
What it is:	Mean, minimum, maximum, and standard deviation of the time gap between consecutive packets in the flow.	
DDoS relevance:	Attack traffic shows very small and uniform inter-arrival times caused by automated packet generation tools whereas benign traffic varies naturally.	
Packet size		
What it is:	Mean, minimum, and maximum packet size observed across all packets in the flow, measured in bytes.	
DDoS relevance:	Attacks often use fixed or very small packet sizes to maximise packet rate whereas benign traffic shows wider size variation.	
Rate		
What it is:	Average bytes per packet, packets per second, and bytes per second computed over the flow duration.	
DDoS relevance:	Captures the high throughput signature that distinguishes flood traffic from normal communication and identifies sustained high-volume attacks.	
Directional ratio		
What it is:	Ratio of forward to backward packet counts and ratio of forward to backward byte counts.	
DDoS relevance:	Benign flows are roughly bidirectional whereas DDoS flows are heavily one-sided with no replies which produces extreme ratios.	
TCP flag		
What it is:	Counts of packets carrying SYN, ACK, FIN, RST, and PSH flags within the flow.	
DDoS relevance:	These map directly to the TCP-flag attack classes which made classification too easy on Run A. Removed in Run B and Run D to force ensemble disagreement.	

Appendix B. Per-Fold Metrics for All Strategies and Runs

Supporting per-fold breakdown for the aggregate results discussed in Section 4.2 (Run A), Section 4.3 (Run B) and Section 4.4 (Runs C and D).

Table 8: Appendix B - Per-fold F1 and aggregate metrics, Run A (27 features, 99,999 flows)

Strategy	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	F1 mean $\pm$ std	MCC	Accu	Prec	Recall
Majority Vote	0.9631	0.9657	0.9619	0.9633	0.9629	0.9634 ± 0.0013	0.9619	0.9653	0.9683	0.9653
Weighted Vote	0.9631	0.9657	0.9619	0.9633	0.9629	0.9634 ± 0.0013	0.9619	0.9653	0.9683	0.9653
Stacking	0.9631	0.9657	0.9619	0.9633	0.9629	0.9634 ± 0.0013	0.9619	0.9653	0.9683	0.9653
Adaptive Vote	0.9631	0.9657	0.9619	0.9633	0.9629	0.9634 ± 0.0013	0.9619	0.9653	0.9683	0.9653

Table 9 Appendix B - Per-fold F1 and aggregate metrics, Run B (22 features, 99,999 flows)

Strategy	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	F1 mean $\pm$ std	MCC	Accu	Prec	Recall
Majority Vote	0.4118	0.4096	0.4096	0.4184	0.4163	0.4131 ± 0.0036	0.3848	0.4386	0.4573	0.4386
Weighted Vote	0.4181	0.4183	0.4174	0.4155	0.4162	0.4171 ± 0.0011	0.3774	0.4359	0.4528	0.4359
Stacking	0.4209	0.4282	0.4262	0.4113	0.4231	0.4219 ± 0.0059	0.3823	0.4393	0.4599	0.4393
Adaptive Vote	0.4181	0.4186	0.4172	0.4155	0.4163	0.4172 ± 0.0011	0.3776	0.4361	0.4529	0.4361

Table 10: Appendix B - Per-fold F1 and aggregate metrics, Run C (27 features, scaled subset)

Strategy	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	F1 mean $\pm$ std	MCC	Accu	Prec	Recall
Majority Vote	0.9636	0.9629	0.9628	0.9631	0.9628	0.9631 ± 0.0003	0.9616	0.9650	0.9699	0.9650
Weighted Vote	0.9636	0.9629	0.9628	0.9631	0.9628	0.9631 ± 0.0003	0.9616	0.9650	0.9699	0.9650
Stacking	0.9634	0.9629	0.9628	0.9631	0.9628	0.9630 ± 0.0002	0.9616	0.9649	0.9698	0.9649
Adaptive Vote	0.9636	0.9629	0.9628	0.9631	0.9628	0.9631 ± 0.0003	0.9616	0.9650	0.9699	0.9650

Table 11: Appendix B - Per-fold F1 and aggregate metrics, Run D (22 features, scaled subset)

Strategy	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	F1 mean $\pm$ std	MCC	Accu	Prec	Recall
Majority Vote	0.4150	0.4188	0.4186	0.4170	0.4202	0.4179 $\pm$ 0.0018	0.3845	0.4405	0.4479	0.4405
Weighted Vote	0.4199	0.4195	0.4178	0.4176	0.4196	0.4189 $\pm$ 0.0010	0.3817	0.4391	0.4583	0.4391
Stacking	0.4231	0.4227	0.4256	0.4217	0.4219	0.4230 $\pm$ 0.0014	0.3859	0.4422	0.4685	0.4422
Adaptive Vote	0.4199	0.4194	0.4177	0.4177	0.4197	0.4189 $\pm$ 0.0010	0.3817	0.4392	0.4583	0.4392

Appendix C. Baseline and Ensemble Leaderboards at Full Precision

Full 7decimal place rankings supporting the 4 decimal place tie discussed in section 5.2.

Table 12: Appendix C - Baseline classifier ranking for Run A by Weighted Performance Score (WPS)

Model	WPS	F1	MCC	Accuracy	Precision	Recall
Decision Tree	0.9641050	0.9633838	0.9619452	0.9652896	0.9701474	0.9652897
Logistic Regression	0.9640843	0.9633638	0.9619226	0.9652696	0.9701274	0.9652697
SVM	0.9640721	0.9633476	0.9619125	0.9652596	0.9701174	0.9652597
MLP	0.9640625	0.9633448	0.9618990	0.9652496	0.9700975	0.9652497
Gradient Boosting	0.9639723	0.9632686	0.9618020	0.9651696	0.9699526	0.9651697
Naive Bayes	0.9628016	0.9621999	0.9605518	0.9640896	0.9684371	0.9640897
Random Forest	0.9626348	0.9621334	0.9603537	0.9639696	0.9678957	0.9639696
KNN	0.9596618	0.9595622	0.9571172	0.9612596	0.9636491	0.9612597

Table 13: Appendix C - Baseline classifier ranking for Run B after feature reduction (22 features)

Model	WPS	F1	MCC	Accuracy	Precision	Recall
Gradient Boosting	0.4189326	0.4248197	0.3807581	0.4400344	0.4600974	0.4400344
Decision Tree	0.4169367	0.4199554	0.3859991	0.4407144	0.4397409	0.4407139
MLP	0.4166439	0.4169480	0.3806019	0.4378644	0.4706563	0.4378634
SVM	0.4109687	0.4085086	0.3810692	0.4312343	0.4586137	0.4312344
Logistic Regression	0.3991609	0.3801919	0.3801314	0.4200642	0.4703831	0.4200643
Naive Bayes	0.3969429	0.3792328	0.3928889	0.4219743	0.4085160	0.4219721
KNN	0.3905526	0.4130118	0.3375640	0.4096541	0.4231578	0.4096540
Random Forest	0.3894165	0.4144807	0.3335470	0.4074541	0.4242062	0.4074542

Table 14: Appendix C - Ensemble Strategy Leaderboard, Run A (27 features, 99,999 flows)

Strategy	F1 mean	F1 std	MCC	Accuracy	Precision	Recall
Majority Vote	0.9633838	0.0013973	0.9619452	0.9652896	0.9701474	0.9652897
Weighted Vote	0.9633738	0.0014105	0.9619339	0.9652796	0.9701374	0.9652797
Stacking	0.9633738	0.0014105	0.9619339	0.9652796	0.9701374	0.9652797
Adaptive Vote	0.9633738	0.0014105	0.9619339	0.9652796	0.9701374	0.9652797

Table 15: Appendix C - Ensemble Strategy Leaderboard, Run B (22 features, 99,999 flows)

Strategy	F1 mean	F1 std	MCC	Accuracy	Precision	Recall
Majority Vote	0.4131324	0.0040134	0.3847795	0.4385644	0.4572820	0.4385636
Weighted Vote	0.4171224	0.0012172	0.3774385	0.4359343	0.4528263	0.4359335
Stacking	0.4219344	0.0065681	0.3822567	0.4392944	0.4599356	0.4392934
Adaptive Vote	0.4171631	0.0012684	0.3776095	0.4360544	0.4528624	0.4360535

Table 16: Appendix C - Ensemble Strategy Leaderboard, Run C (27 features, scaled subset)

Strategy	F1 mean	F1 std	MCC	Accuracy	Precision	Recall
Majority Vote	0.9630542	0.0003071	0.9616302	0.9649956	0.9699265	0.9649956
Weighted Vote	0.9630542	0.0003071	0.9616302	0.9649956	0.9699265	0.9649956
Stacking	0.9630104	0.0002354	0.9615751	0.9649496	0.9698489	0.9649496
Adaptive Vote	0.9630542	0.0003071	0.9616302	0.9649956	0.9699265	0.9649956

Table 17: Appendix C - Ensemble Strategy Leaderboard, Run D (22 features, scaled subset)

Strategy	F1 mean	F1 std	MCC	Accuracy	Precision	Recall
Majority Vote	0.4179047	0.0017989	0.3844916	0.4404904	0.4479196	0.4404904
Weighted Vote	0.4188843	0.0009707	0.3816554	0.4391484	0.4582955	0.4391484
Stacking	0.4229897	0.0014003	0.3858888	0.4421664	0.4684865	0.4421664
Adaptive Vote	0.4188949	0.0009791	0.3816601	0.4391844	0.4582827	0.4391844

Appendix D. Per-Fold Disagreement Across All Four Runs and Strategies

Per-fold disagreement data supporting the heterogeneity-collapse finding in Section 5.1 and Sec 6.1.

Table 18: Appendix D - Pairwise disagreement averaged (%) across five cross-validation folds for all four experimental

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean	Range
Run A	0.0000%	0.0000%	0.0100%	0.0100%	0.0100%	0.0060%	0.0000-0.0100%
Run B	39.7200%	41.6500%	39.5200%	34.6500%	39.5300%	39.0140%	34.6500-41.6500%
Run C	0.0525%	0.0028%	0.0032%	0.0018%	0.0025%	0.0126%	0.0018-0.0525%
Run D	37.9500%	37.3200%	37.7200%	38.0900%	37.8200%	37.7820%	37.3200-38.0900%

Appendix E. AI Transparency Statement (AITS 2)

AITS Descriptor: AITS 2 -AI for Shaping

AI Tools Used: Perplexity AI

How AI Was Used:

I used Perplexity to help break down the assignment brief into sections to help clarify what the assignment requirements were and to help understand tasks.

- Used AI to explain the assignment criteria for a First
- Used Ai to help suggest structure for the report for example (Recommended Headings from Steve + Subsection title recommendations from Ai)
 - Used Ai to minimal code assisting
- Used Ai for light grammar recommendations due to neurodiversity

(All writing, technical analysis, interpretation of results are based on my own understanding)

Appendix F. Git Hub Repository

Source Code Repository (for reproducibility):

[DanKinsiona/ddos-detection-heterogeneous-ensembles: Heterogeneous ensemble methods for DDoS detection. BSc dissertation code supporting four-run comparative study \(27/22 features × baseline/scaled\) of Majority Vote, Weighted Vote, Stacking, and Adaptive Voting on a 9-class flow-level dataset. Sheffield Hallam University, 2026.](#)

<https://github.com/DanKinsiona/ddos-detection-heterogeneous-ensembles>